Kayané ROBACH

robachowyk@gmail.com +(31) 6 26 57 65 73 +(33) 7 81 94 78 77 https://kayanerobach.github.io/

WORK EXPERIENCE

2022 - 2026 Consultant in Statistics; Amsterdam UMC, The Netherlands

Statistical consultancy (data modelling, sampling methods, causal discovery, hypothesis testing) for medical researchers.

2022 - 2026 Teaching Assistant; Vrije Universiteit Amsterdam, Amsterdam UMC Doctoral School, The Netherlands

Leading exercise classes and computer practical sessions for Bachelor students for the course: Statistics for Medical Research into Clinical Practice and for PhD candidates for the course: Computing in R. (SPSS, R, Python)

2022 Research Assistant; LIGM Paris, France

Research project on the feasibility and stability properties of the equilibria in Lotka Volterra models for theoretical ecology in large dimension. Implementation of an algorithm for phase transition detection. (MatLab, Python)

2021 Data Scientist, Research & Development; 55 Paris, France

R&D projects on Reinforcement Learning, Agent Based Models, and Cognitive Science for marketing. Implementation of simulation studies, AB-Tests, evaluation surveys to ameliorate marketing Key Performance Indicators. (SQL, Python)

2020 Data Scientist; EvidenceB Paris, France

Implementation of data processing pipelines, data analysis, clustering and recommendation algorithms on the launch of a new interactive online learning tool using AI at the service of school curriculums. (Python)

EDUCATION

2022 - 2026 PhD in Mathematical Statistics; Amsterdam UMC, The Netherlands

Causal Record Linkage *Developed novel statistical methods for causal inference and data fusion. Implemented statistical models for record linkage of heterogeneous data sources. Participated to international conferences (ICSDS, Eurocim, CompStats, IBS, IBC). 2 published articles (JRSSSC, Statistics in Medicine) and 2 articles in submission (R journal, Journal of Causal Inference). Automation of software maintenance with agentic coding (lamma-cpp, opencode) (Python, R, C++)*

Department Outing Committee *Organisation of the Department of Epidemiology and Data Science annual outing.*

2021 - 2022 Master 2 Mathematics and Applications; ENS Paris-Saclay, France

MVA Relevant technical projects: *architecture design, implementation, training of models for object recognition & computer vision (CNN, bag-of-features, gradient based learning); of advanced learning methods for text and graph data (NLP, graph-based text mining, knowledge distillation); of generative models, deep reinforcement learning models, RNN, LSTM, GAN, transformers. (PyTorch, Keras, TensorFlow)*

2020 - 2021 Master 1 Applied Mathematics and Statistics; Toulouse School of Economics, France

Mathematics and Economic Decision *Games and equilibria, Probability modelling, Strategic and dynamic optimisation, Time series, Martingales theory, Advanced functional analysis*

2019 - 2022 Magister Econometrics and Statistics; Toulouse School of Economics, France

Economist Statistician *Machine Learning, Stochastic simulation, Markov chains and applications, Data analysis, Programming, Statistics, Optimisation for big data*

2017 - 2020 Double Degree Economics and Mathematics; Toulouse School of Economics, France

Economics, Mathematics and Computer Science applied to human and social sciences *Statistics, Measure theory and probability, Algebra, Algorithmic, Micro and Macro Economics, Analysis and optimisation, Econometrics*

OUTREACH

2025 - now CoMeEcon Seminar; VVS-OR, The Netherlands *Manage funds, program operations, communications and logistics of the biannual seminar, funded by the VVS-OR mathematical statistics section, on computational statistics.*

2024 - now Fietsenwerkplaats Smerig; Amsterdam, The Netherlands *Volunteer work; build and repair bicycles; assemble and disassemble bikes, wheels, internal hub mechanisms.*

2017 - 2022 Student support & Guidance; Toulouse, Paris, France *Educational support for children with scholar difficulties and for young students.*

MISCELLANEOUS

Tools: R, C++, SAS, STATA, SQL, Python (PyTorch, Keras, TensorFlow), llama.cpp, opencode
French (native), English (fluent), Dutch (intermediate), Spanish (intermediate)
Piano & Violin amateur player, amateur sailor